Applied Corporate Finance

Week #3C – Capital Structure

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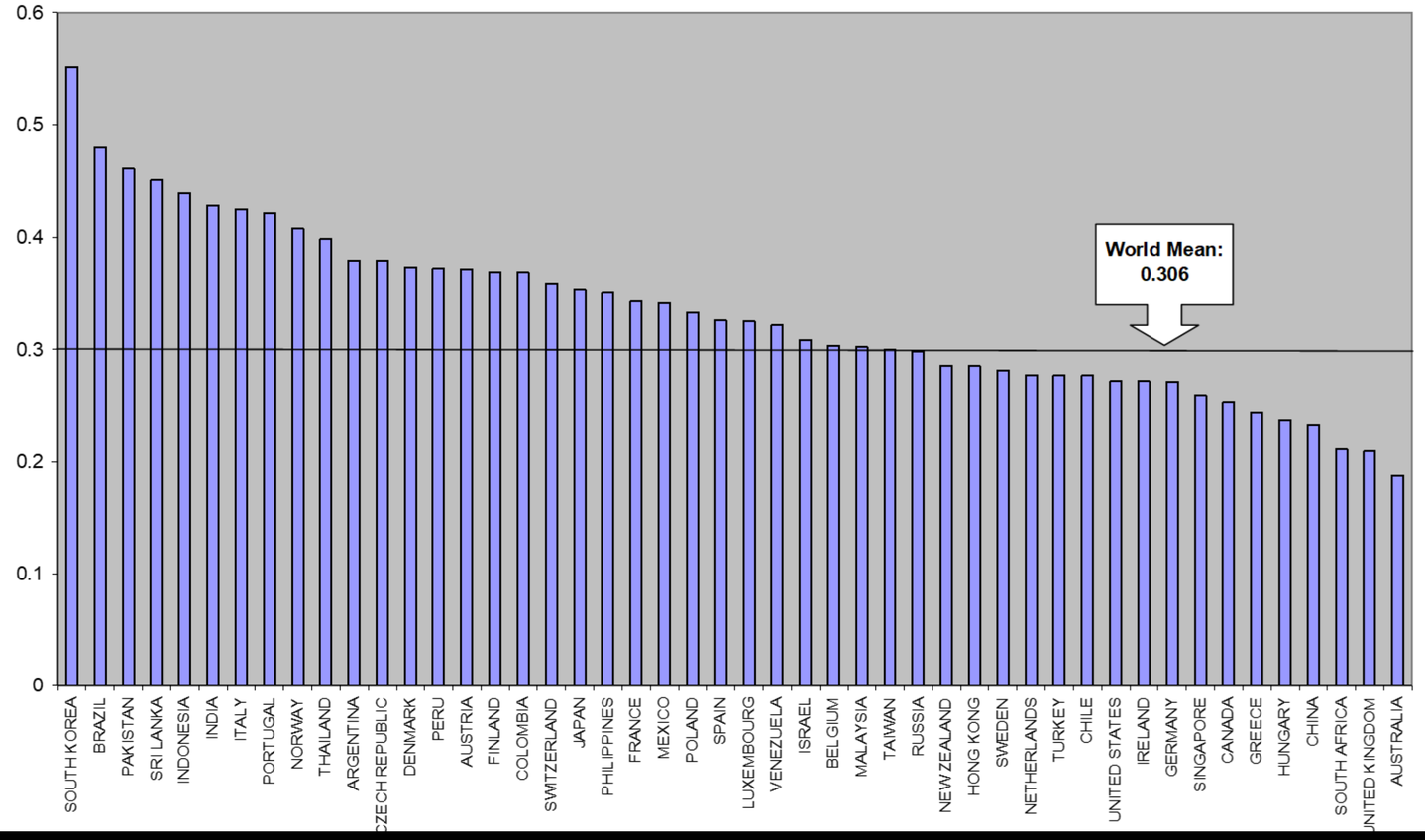
Plan of Attack

- **Corporate Capital Structure: Some Stylized Facts**

- **Theories of Capital Structure – Traditional and Modern**

Capital Structure: Some Stylized Facts

Leverage Ratios Around the World: 1990-2005



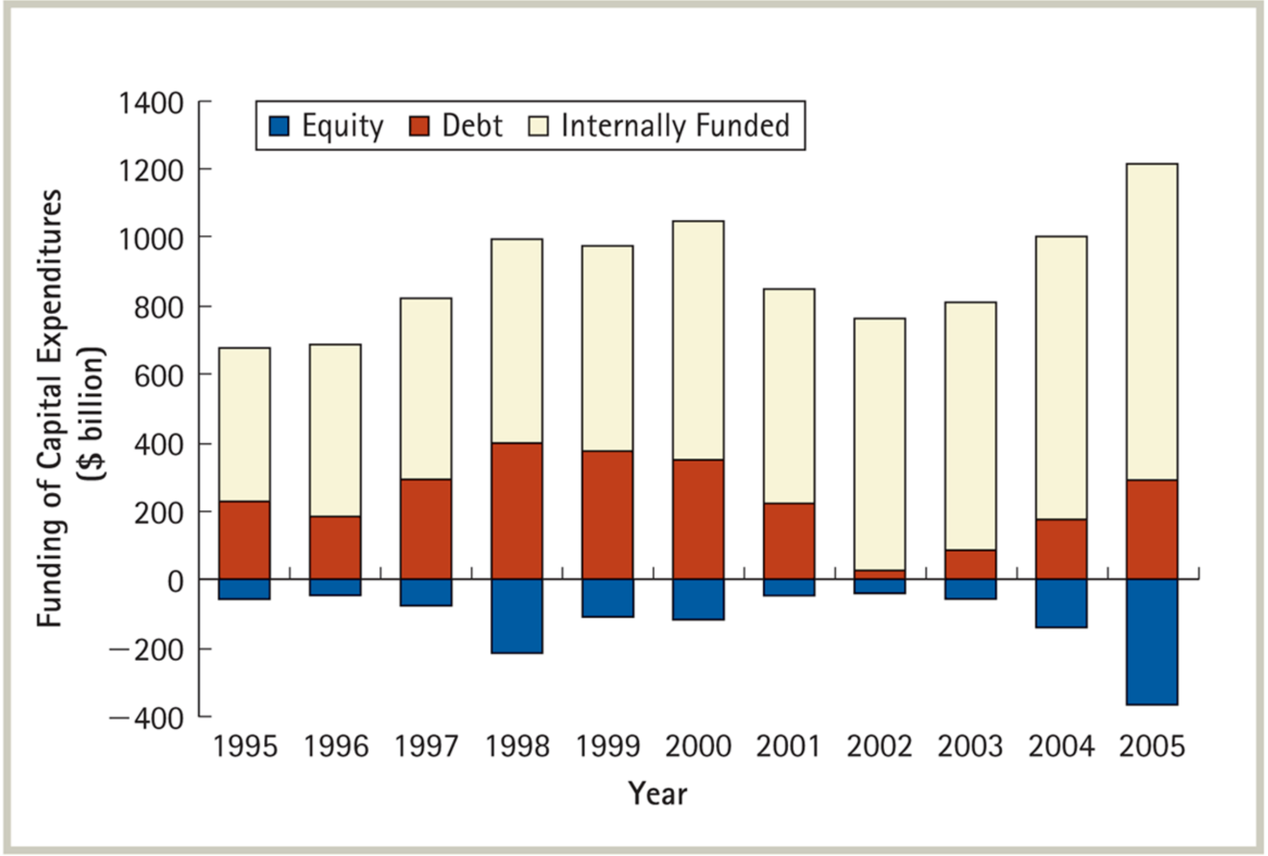
Source: Brau & Fawcett (2006)

Stylized Fact #1

- There is significant variation in average leverage ratios across countries.

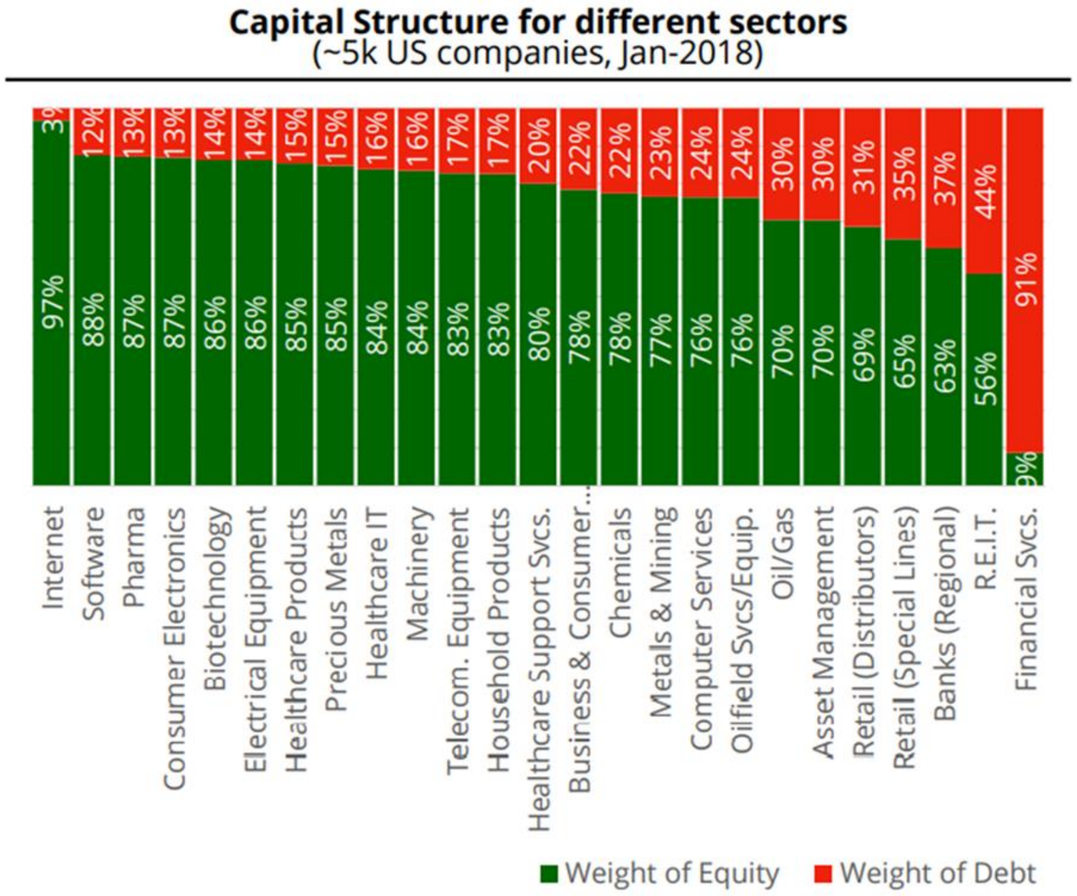
- What might explain these differences?
  - Legal origin (Common Law vs. Civil Law) – (Common Law, better investor protection).
  - Relative size of the banking sector
  - Tax system (treatment of dividends)
  - Bankruptcy code
  - Power of labour unions
  - Degree of economic development
  - Government debt guarantees

Stylized Fact #2



- Internally generated funds are the primary source of financing, followed by debt

Stylized Fact #3



- There is significant variation in average leverage ratios across industries.

Stylized Fact #4

- 6 factors are consistently correlated with cross-sectional differences in leverage:

Factor	Relationship
Fixed Assets (tangibility of assets)	**+**
Firm Size	**+**
Time since IPO	**+**
Growth opportunities	**-**
Profits	**-**
Dividends Payer	**-**

Stylized Fact #5

- Corporate leverage is mean-reverting at the firm level.
  - Empirical studies estimate target adjustment rates at 8-10% per year.
  - Firms often have a target leverage ratio, or target credit rating.

Stylized Fact #6

- Approximately 70% of firms in a given year issue some amount of equity
  - This is much more than would be expected if firms strictly followed the pecking order theory.

- After an IPO, equity issues are more important for small firms than for large firms.
  - Large firms tend to issue equity less frequently, but on a larger scale.

Stylized Fact #7

Composition of Debt:

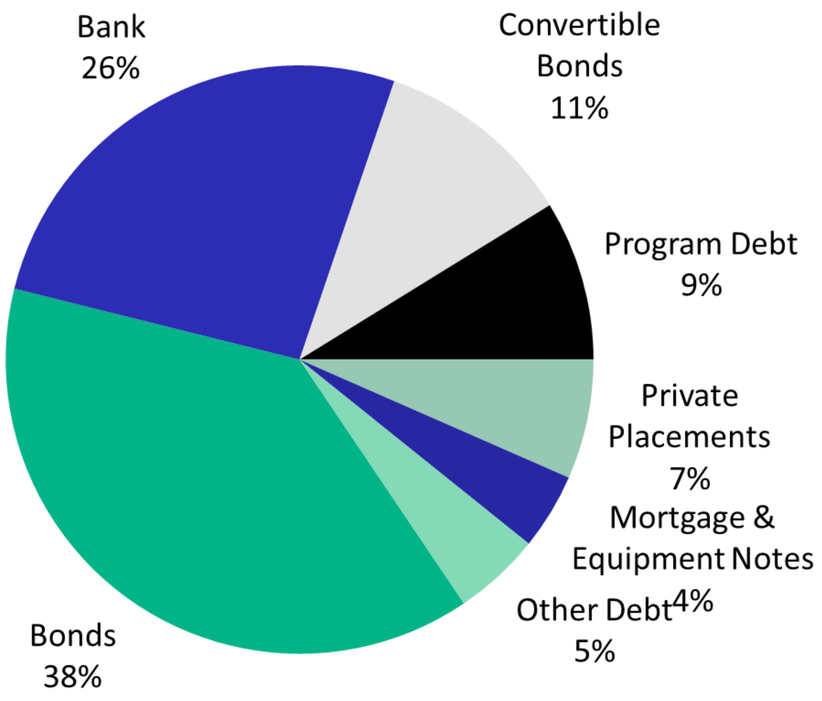
Rated US Firms

DEBT	Share of capital Debt Type/(D+E)	Share of Total Debt Debt Type/(Total Debt)
Total Debt	**50,2%**	**100,0%**
Bonds	**19,2%**	**38,2%**
Bank Loans	**13,2%**	**26,3%**
Convertible Bonds	**5,5%**	**11,0%**
Program Debt	**4,4%**	**8,8%**
Private Placements	**3,3%**	**6,6%**
Motgage Debt & Equipment	**2,1%**	**4,2%**
Other	**2,4%**	**4,8%**

- Program Debt: Commercial paper, MTNs, shelf-registered debt.
- Other: Capitalized leases, unclassified debt.

Stylized Fact #7 (cont.)

Composition of Debt For Rated Firms 1996-2006



Theories of Capital Structure

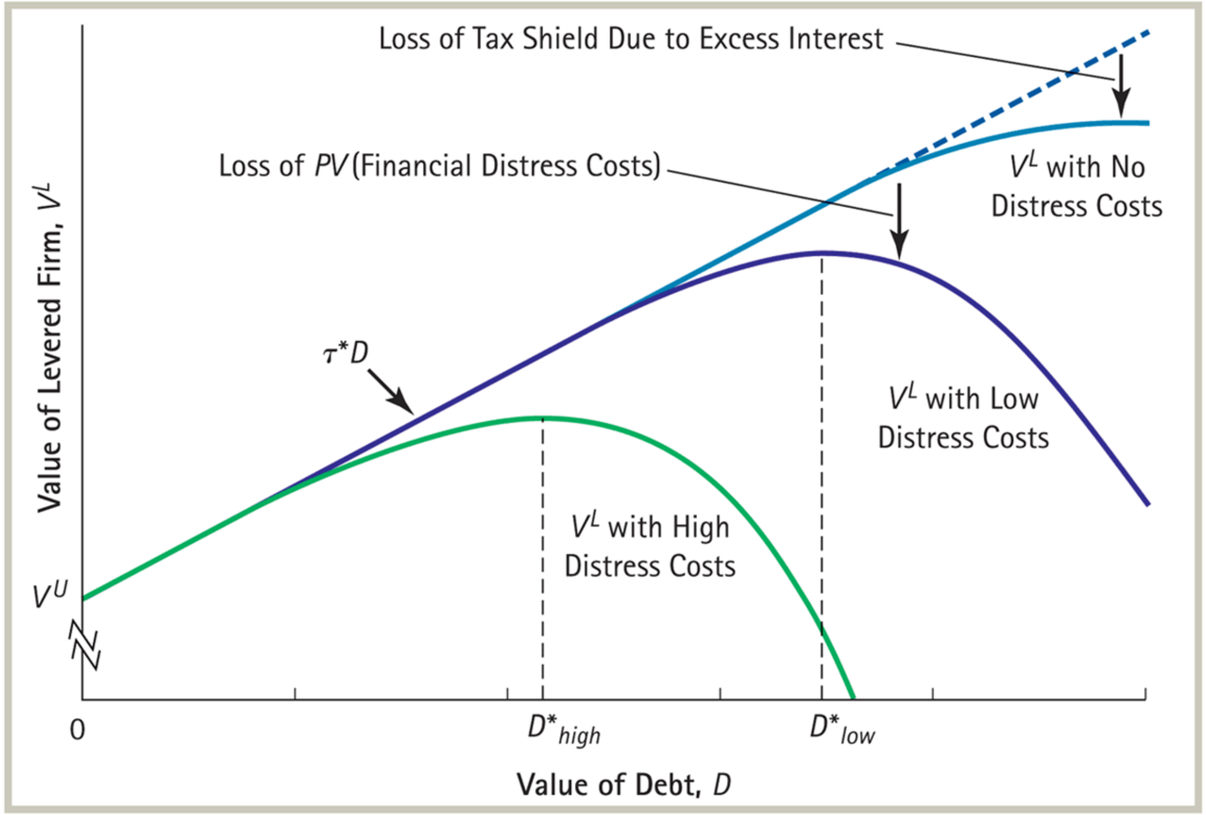
Static Trade-off Theory

- Static Model (Modigliani and Miller) Predicts:

$$V^L = V^U + PV(\text{Interest Tax Shield}) - PV(\text{Financial Distress Costs})$$

  - An increase in non-debt tax shields reduces optimal leverage
  - Increase in the tax on interest income (o lenders) decreases debt
  - Increase in personal tax on equity increases leverage

Static Trade-off Theory (cont.)



Costs of Financial Distress

		Examples
Direct costs	Legal expenses Court costs Advisory fees (...) Other	• When facing financial distress and close to or in bankruptcy the company will have extra expenses that it would not have otherwise if it did not have debt. Increasing debt will increase these costs and/or the probability of occurrence
Indirect costs	Reputation	• Financial distress may affect demand, since the firm cannot commit to remain in business. Examples: (A) Products which may require spare parts and maintenance (e.g. automobiles), (B) Firms which depend on customer relationships (e.g. airlines) • Suppliers may stop sending raw materials • Firms lose flexibility when so closely monitored by creditors • Employees may be less willing to make firm-specific investments when the firm is in or near distress
	Fire sales	• Assets may be liquidated at fire sale prices because (A) Creditors have little incentive to search for prices higher than the value of their claims or (B) Firms with highly specific assets that cannot be sold quickly without significant losses in value will suffer most (e.g. oil drilling)

Considering Agency Costs

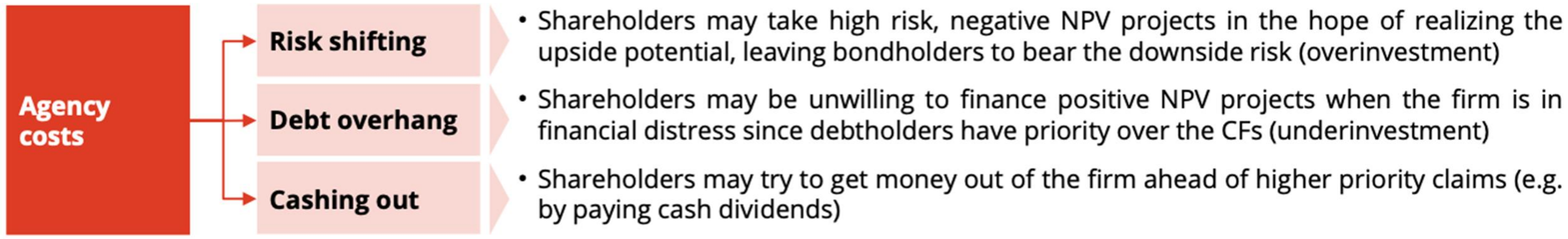
$$V^L = V^U + PV(\text{Interest Tax Shield}) - PV(\text{Financial Distress Costs}) + PV(\text{Agency Benefits}) - PV(\text{Agency Costs})$$

Agency Benefits of Debt (prevents):

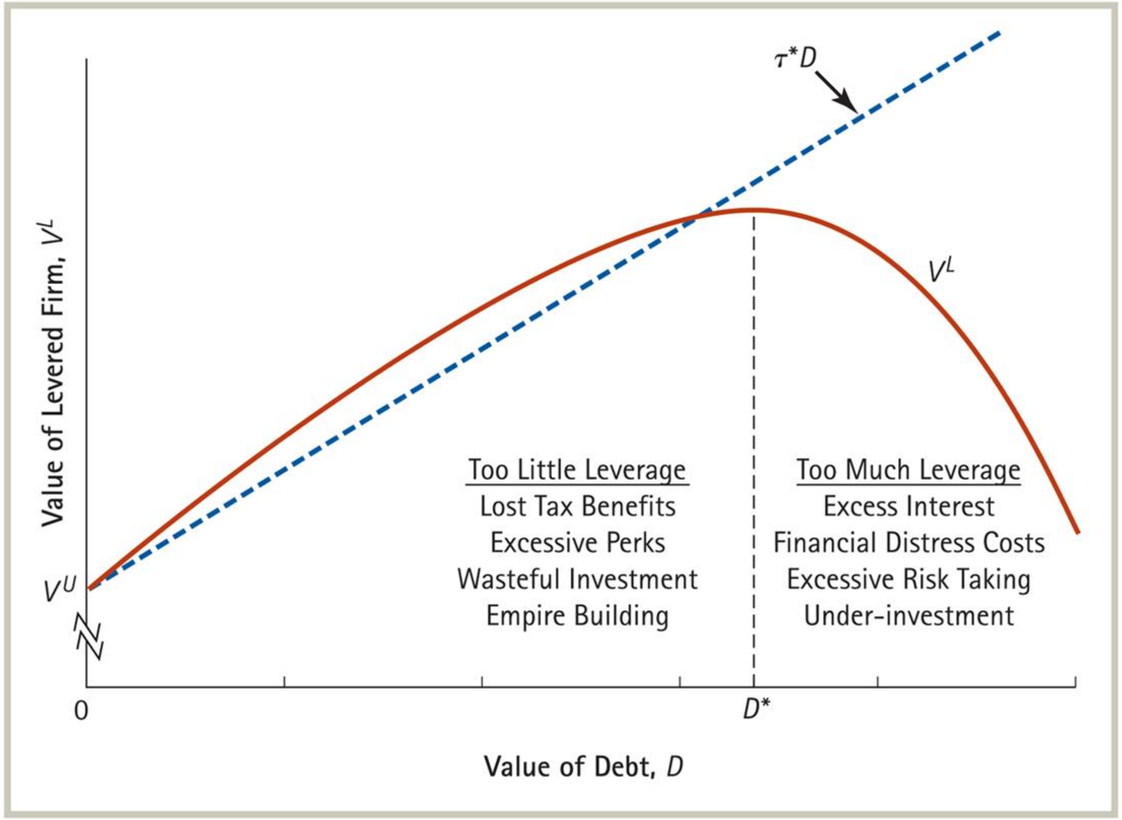
- Managerial perquisites
- Empire Building

Agency Costs of Debt:

- Risk-Shifting (asset substitution)
- Underinvestment (debt overhang)
- Cashing out



Trade-off theory with Agency Costs/Benefits



Pecking Order Hypothesis

1. Pecking Order Theory (Myers, 1984; Myers & Majluf, 1984) is developed under high information asymmetries situations, namely between managers and external investors.

2. Firms prefer internal to external finance.
   - Note: Dividends are “sticky,” so dividend cuts are not made to finance capital expenditure.

3. If external finance is required, firms will issue the safest (i.e., lowest information sensitivity) security first
   - debt (long term) before equity.
   - firms issue equity as a last resort.
   - Why is equity the most “informationally sensitive”?

Trade-off Static vs Pecking Order Hypothesis

1. Shyam-Sunder and Myers (1999) and Frank and Goyal (2003) presented empirical tests on the pecking order and trade-off theories in the context of U.S. listed firms. They concluded that these firms did not follow pecking order theory.

2. Fama and French (2002) used the same data as Shyam-Sunder and Myers to test pecking order theory and found evidence of a negative relationship between profitability and leverage , which they proved to be consistent with pecking order theory.

3. Lemmon and Zender (2010) demonstrate small and high-growth firms' preference for equity finance based on a sample of US listed firms between 1971 and 2002, attributing this to their growth levels and restrictive debt capacity constraints.

4. We should expect to verify the pecking order theory in nascent firms, startups and small and medium innovative firms, firms with more information asymmetries, but the evidence is contrary, these firms are financed mainly by equity in their first years. Some authors speculate that promoter's equity may somehow be considered as internal funds of these firms, as sometimes it is difficult to distinguish between companies and promoters' funds.

5. There has been some evidence of pecking-order theory in family-owned firms.

Debt Conservatism Puzzles

- Low-Leverage Puzzle:
  - Graham (2000) finds that firms are under-levered with respect to their marginal tax rates.
  - Estimates that firms could add an average of 7.3% to firm value by increasing leverage to the optimal level.

- Zero-Leverage Puzzle:
  - Approximately 10% of firms have no (or even negative) leverage.
  - 23% of firms have leverage ratios less than 0.05.
  - This is not just a small firm phenomenon. Why?

Dynamic Models and Debt Conservatism

- In dynamic models, the firm must come to the capital markets for financing from time to time.

- The cost of external finance varies over time according to macroeconomic conditions and the firm's financial strength.
  - E.g., Erel, Julio, Kim, and Weisbach (2009): **Firms with higher probability of default** (high leverage, low rating) are **rationed out of debt markets during market downturns**.
- Firms that have a large proportion of value coming from growth options will choose financial structures that avoid the risk of losing access to external finance.

- The benefit from maintaining access to finance will, in some cases, out-weight the tax benefits of debt for growth firms – even if they are profitable

  ⇒ firms will appear under-levered relative to a static trade-off model.

Behavioural Finance and Capital Structure

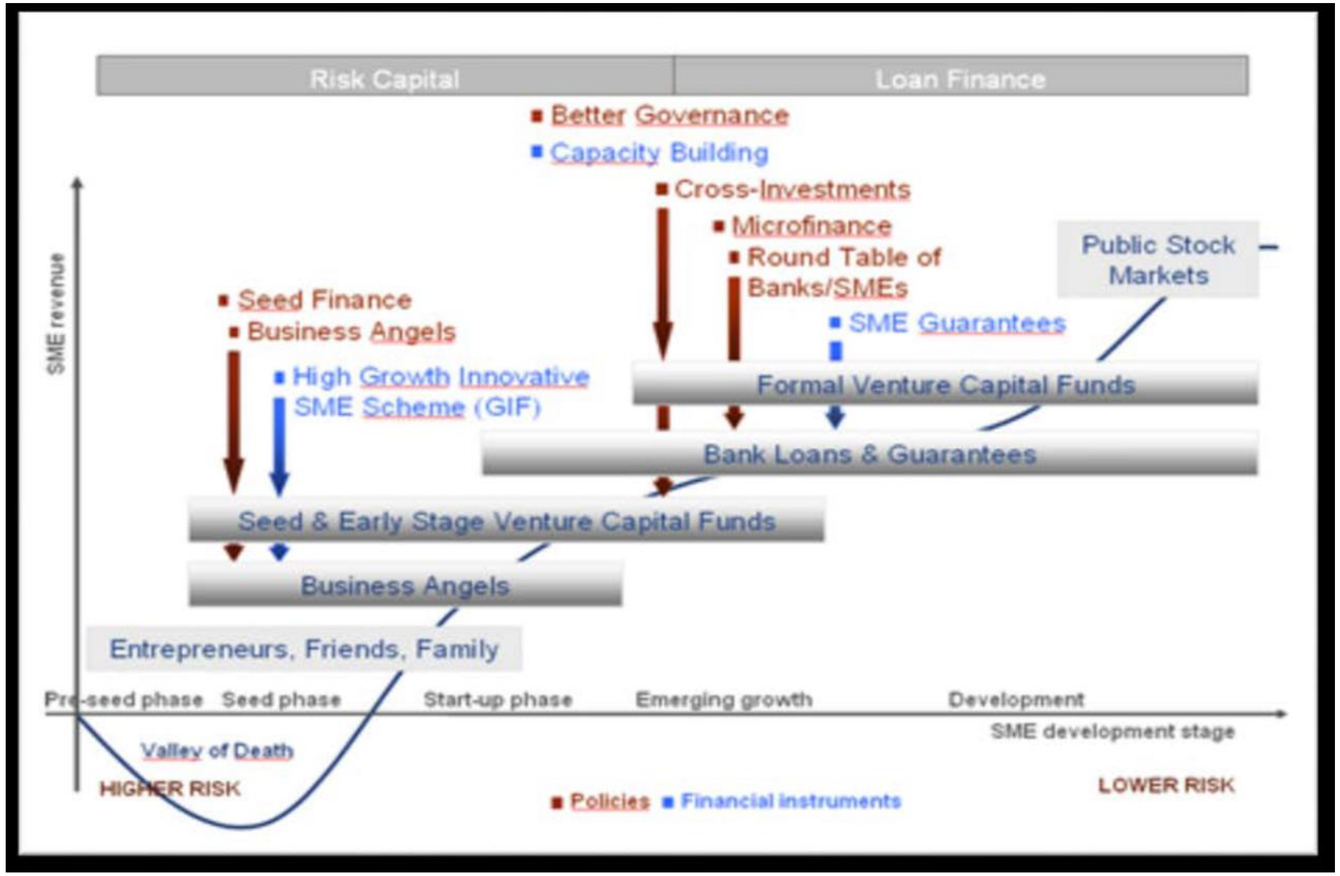
1. Irrational Managers

- **Overconfidence/Overoptimistic:**
  - Overconfident managers will have otherwise higher leverage ratios as they over-estimate cash flows and under-estimate volatility.
- **Overweighting Personal Experience:**
  - Graham and Narasimhan (2004) find that firms with managers who lived through the great depression have significantly lower leverage ratios.

2. **Irrational Investors**

- Rational managers take advantage of the mis-pricing of equity and issue when the stock price is irrationally high.
- Explains: price run-up, negative price reaction, long-run underperformance of SEOs
- Implies that a firm's observed capital structure is the cumulative effect of attempts to time the market.

SMEs life cycle and financing needs



Summary: Capital Structure

- The goal of capital structure is to choose the financing mix that maximizes the value of the firm.

- Finance can add/subtract value in places where the Modigliani-Miller assumptions fail. For example,
  - **Tax shield benefits**
  - **Costs of financial distress**
  - **Agency problems**
    - Mitigate FCF problem?
    - Cause debt overhang/risk shifting problems?